Math 67 – HW 5 Solutions

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Chapter 6

C1. Map $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x, y) = (x + y, x)$.

(d) We want matrix for T with respect to the canonical basis for both the domain and codomain (which are both $\mathbb{R}^2$). The canonical (standard) basis for $\mathbb{R}^2$ is $\{e_1, e_2\}$ where $e_1 = (1, 0)$ and $e_2 = (0, 1)$. We first evaluate T on the basis for the domain. $T(e_1) = (1, 1)$ and $T(e_2) = (1, 0)$. Then, express them as a linear combination of basis vectors of the codomain, which in this case are also e_1 and e_2. This is done by solving the equation

$$a_{11}e_1 + a_{21}e_2 = T(e_1) \quad \Leftrightarrow \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad a_1 = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Similarly, we have

$$Ia_2 = T(e_2) = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

so $M(T) = \begin{bmatrix} a_1 & a_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. Note that in this case (canonical bases for $\mathbb{R}^n$) we can use the shortcut

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x & + & y \\ x & & \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}}_{M(T)} \begin{bmatrix} x \\ y \end{bmatrix}.$$

(e) This time, canonical basis $\{e_1, e_2\}$ for the domain and $\{v_1 = (1, 1), v_2 = (1, -1)\}$ for the codomain:

$$a_{11}v_1 + a_{21}v_2 = T(e_1) \quad \Leftrightarrow \quad \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad a_1 = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} a_2 = T(e_2) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \Rightarrow \quad a_2 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix},$$

and then $M(T) = \begin{bmatrix} 1 & 1/2 \\ 0 & 1/2 \end{bmatrix}$.

C2. Map $T \in \mathcal{L}(\mathbb{R}^2)$ defined by $T(x, y) = (y, -x)$.
(c) Canonical bases for both domain $\mathbb{R}^2$ and codomain $\mathbb{R}^2$, so

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} & y \\ -x & \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}}_{M(T)} \begin{bmatrix} x \\ y \end{bmatrix}.$$

P8. Let V be a finite dimensional vector space over $\mathbb{F}$ with $S, T \in \mathcal{L}(V, V)$. Then $T \circ S$ is invertible if and only if both S and T are invertible.

Proof. Since S,T, and $T \circ S$ are endomorphisms (see text p. 68), Theorem 6.7.6 implies that for these three maps, invertibility, injectivity, and surjectivity are equivalent properties. We will make use of this fact.
($\Rightarrow$) Recall from chapter 6 #3 (it was on last week's hw) that if S and T are injective then $T \circ S$ is injective.
($\Leftarrow$) Now, assume $T \circ S$ is invertible. Then it is injective and surjective. From injectivity, it follows that $\text{null}(T \circ S) = \{0\}$. Since $\text{null}(S) \subseteq \text{null}(T \circ S)$, it follows that $\text{null}(S) = \{0\}$. Thus, S is injective. Surjectivity of $T \circ S$ implies that $\text{range}(T \circ S) = V$. But $\text{range}(T \circ S) \subseteq \text{range}(T)$, so $V \subseteq \text{range}(T)$. It's trivial that $\text{range}(T) \subseteq V$, so we have that $\text{range}(T) = V$ and thus T is surjective. □

P9. Let V be a finite-dimensional vector space over $\mathbb{F}$ with $S, T \in \mathcal{L}(V, V)$, and I is the identity map on V. Then

$$T \circ S = I \quad \Longleftrightarrow \quad S \circ T = I.$$

Proof. ($\Rightarrow$) Assume $T \circ S = I$, and let $x \in V$. Then $T(S(x)) = Ix = x$. Since I is invertible it follows from #8 (the previous hw problem) that S and T are invertible. Then $S(x) = T^{-1}(x)$, and $S^{-1}(x) = T(x)$, and then $x = S(T(x))$. Thus $S \circ T = I$.
($\Leftarrow$) Assume $S \circ T = I$, and let $x \in V$. Then $S(T(x)) = x$. Since I is invertible it follows that S and T are invertible. Then $T(x) = S^{-1}(x)$, and $T^{-1}(x) = S(x)$, and then $x = T(S(x))$. Thus $T \circ S = I$. □

Chapter 7

C1. Define $T \in \mathcal{L}(\mathbb{F}^2)$ as $T : (u, v) \mapsto (v, u)$.
By inspection, one eigenpair is $1 \in \mathbb{F}$ and $(1, 1) \in \mathbb{F}^2$, since $T(1, 1) = 1(1, 1)$.
Also, $-1 \in \mathbb{F}$ and $(1, -1) \in \mathbb{F}^2$.
The two eigenvectors $\{(1, 1), (1, -1)\} \subset \mathbb{F}^2$ are linearly independent and thus span $\mathbb{F}^2$, so this is a *full set* of eigenvectors for T.

C3. For $n \in \mathbb{Z}_+$, define the map $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^n)$ to be

$$T(x_1, x_2, \ldots, x_n) = (s, s, \ldots, s), \quad \text{where} \quad s = x_1 + x_2 + \cdots + x_n.$$

This map has two eigenvalues:
First, $T(x_1, x_2, \ldots, x_n) = s(1, 1, \ldots, 1)$, so for $(x_1, x_2, \ldots, x_n) = (1, 1, \ldots, 1)$ we have

$$T(1, 1, \ldots, 1) = n(1, 1, \ldots, 1),$$

so n is an eigenvalue of T and $(1, 1, \ldots, 1)$ is the associated eigenvector.
Zero is also an eigenvalue, since

$$T(x_1, x_2, \ldots, x_n) = 0(x_1, x_2, \ldots, x_n) = (0, 0, \ldots, 0)$$

for any vector in the set

$$\{ (x_1, x_2, \ldots, x_n) \mid x_1 + x_2 + \cdots + x_n = 0 \} = \text{null}(T),$$

i.e. the eigenspace (span of eigenvectors) associated with the eigenvalue 0 is the nullspace of T. We will determine a basis for null(T): note that any vector $(x_1, x_2, \ldots, x_n) \in \text{null}(T)$ satisfies $x_1 = -x_2 - \cdots - x_n$, so

$$\begin{array}{ccccccc} x_1 & = & -x_2 & -x_3 & \cdots & -x_n \\ x_2 & = & x_2 & & & \\ x_3 & = & & x_3 & & \\ & \vdots & & & & \\ x_n & = & & & & x_n \end{array}$$

Then $(x_1, x_2, \ldots, x_n) \in \text{null}(T) \subset \mathbb{F}^n$ can be expressed as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \cdots + x_n \begin{bmatrix} -1 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

The $n-1$ vectors $(-1, 1, 0, \ldots, 0), (-1, 0, 1, \ldots, 0), \ldots, (-1, 0, 0, \ldots, -1)$ form a basis for null(T), which is the eigenspace associated with the eigenvalue 0. We can call these vectors the eigenvectors of T, associated with 0, but actually any $v \in \text{null}(T)$ is an eigenvector, so in this case (eigenvalue with multiple eigenvectors) it makes more sense to talk about the eigenspace and its basis.

P2. Let V be a finite-dimensional vector space over $\mathbb{F}$ with $T \in \mathcal{L}(V, V)$ and suppose that U_1 and U_2 are subspaces of V which are invariant under T. Then $U_1 \cap U_2$ is also an invariant subspace of V under T.

Proof. The set $U_1 \cap U_2$ a subspace of V since U_1 and U_2 are subspaces of V (this was shown in a previous homework). Let $w \in U_1 \cap U_2$. Then $w \in U_1$ and since U_1 is invariant under T, we have $Tw \in U_1$. Similarly, $w \in U_2$ where U_2 is T-invariant implies that $Tw \in U_2$. Thus $Tw \in U_1 \cap U_2$ and it follows that $U_1 \cap U_2$ is invariant under T. □

P4. Let V be a finite-dimensional vector space over $\mathbb{F}$ and $T \in \mathcal{L}(V, V)$ has the property that every $v \in V$ is an eigenvector for T. Then T is a scalar multiple of the identity function on V.

Proof. Since every vector in V is an eigenvalue of T, let $\{v_1, v_2, \ldots, v_n\}$ be a basis for V, and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the assoicated eigenvalues, so that $Tv_j = \lambda_j v_j$ for $j = 1, 2, \ldots, n$.
Let $x \in V$, and let $\mu \in \mathbb{F}$ be the eigenvalue associated with x. For some fixed $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{F}$, express x as $x = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$, so that

$$\begin{aligned} Tx &= \alpha_1 Tv_1 + \alpha_2 Tv_2 + \cdots + \alpha_n Tv_n \\ &= \alpha_1 \lambda_1 v_1 + \alpha_2 \lambda_2 v_2 + \cdots + \alpha_n \lambda_n v_n \end{aligned}$$

and

$$\mu x = \alpha_1 \mu v_1 + \alpha_2 \mu v_2 + \cdots + \alpha_n \mu v_n.$$

Then we must have $\lambda_1 = \lambda_2 = \cdots = \lambda_n = \mu$, and thus it is apparent that μ is the eigenvalue associated with every vector in V, i.e. for any $v \in V$,

$$Tv = \mu v,$$

and it follows that $T = \mu I$. □

P5. Let V be a finite-dimensional vector space over $\mathbb{F}$, and let $S, T \in \mathcal{L}(V)$ be linear operators on V with S invertible. Given any polynomial $p(z) \in \mathbb{F}[z]$,

$$p(S \circ T \circ S^{-1}) = S \circ p(T) \circ S^{-1}.$$

Proof. Recall from p. 64 of the textbook that the composition $S \circ T \circ S^{-1}$ is equivalent to the product STS^{-1}. Observe that for $j \in \mathbb{Z}_+$ due to cancellation of inverses ($S^{-1}S = I$),

$$(STS^{-1})^j = STS^{-1}STS^{-1} \cdots STS^{-1} = ST^jS^{-1}.$$

Then

$$p(STS^{-1}) = \sum_j \alpha_j (STS^{-1})^j = \sum_j \alpha_j ST^jS^{-1} = S\left(\sum_j \alpha_j T^j\right) S^{-1} = Sp(T)S^{-1},$$

where n is the degree of the polynomial and the index $j = 0, 1, 2, \ldots, n$, and $\alpha_j \in \mathbb{F}$ are the coefficients of the polynomial p. □

Note: Both $S \circ T \circ S^{-1}$ and STS^{-1} mean the linear map evaluated as $S(T(S^{-1}(x)))$, for $x \in V$, and $T^3x = TTTx = T(T(T(x)))$.

P6. Let V be a finite-dimensional vector space over $\mathbb{C}$, let $T \in \mathcal{L}(V)$ be a linear operator, and let $p(z) \in \mathbb{C}[z]$ be a polynomial. Then $\lambda \in \mathbb{C}$ is an eigenvalue of the operator $p(T) \in \mathcal{L}(V)$ if and only if T has an eigenvalue $\mu \in \mathbb{C}$ such that $p(\mu) = \lambda$.

Proof. For $p \equiv 0$, this is trivial, so we will assume p is not the zero polynomial.

($\Rightarrow$) Assume $\lambda \in \mathbb{C}$ is an eigenvalue of $p(T)$. Then for some nonzero $v \in V$, we have $p(T)v = \lambda v$. Define the polynomial $q(z) = p(z) - \lambda$, so that $q(T)v = (p(T) - \lambda I)v = 0$. By the Fundamental Theorem of Algebra, we can factor $q(z)$ as

$$q(z) = c(z - \mu_1)(z - \mu_2) \cdots (z - \mu_n),$$

where $c, \mu_j \in \mathbb{F}$ and $c \neq 0$. Then we can write

$$q(T)v = (p(T)v - \lambda I)v = c(T - \mu_1 I)(T - \mu_2 I) \cdots (T - \mu_n I)v = 0.$$

Since $v \neq 0$, at least one of the operators $T - \mu_j I$ must be non-injective, i.e. for some j, the scalar $\mu = \mu_j$ is an eigenvalue of T. Since μ is a root of $q(z)$, we have $q(\mu) = p(\mu) - \lambda = 0$, so $p(\mu) = \lambda$.

($\Leftarrow$) For this direction, we assume T has an eigenvalue $\mu \in \mathbb{C}$ and $\lambda = p(\mu)$. Then

$$p(T)v = \left(\sum_j \alpha_j T^j\right) v = \sum_j \alpha_j T^j v = \sum \alpha_j \mu^j v = p(\mu)v = \lambda v,$$

where $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{F}$ are the coefficients of p $\square$

Note: For the "$\Rightarrow$" direction of the proof, we used an idea from the proof of Theorem 7.4.1, which is one way to show the existence of an eigenvalue.